

Soft Spaces - Phase 4 Status Report

Applications, robustness and Qiskit/Aer circuit-level validation

Status: 15 September 2026

Executive status. Phase 4 has moved from a purely structural Phase-3 result to a concrete application test. The current evidence supports a reproducible REAL-specific advantage for cross-energy ranked search in the frozen 8-qubit model. The advantage has survived classical score noise, a hardware-oriented finite-shot/readout/depolarizing proxy, and an explicit noisy Qiskit/Aer circuit channel through 400 CX gates. The current open question is not whether an application advantage exists in this model, but where it breaks down and whether the same advantage survives on real IBM quantum hardware.

1. Phase 4 objective

Phase 4 asks whether the structure identified in Phase 3 can be used operationally. The first application is cross-energy ranked search: information from the negative-energy partner (-E) is used as a prior to rank candidates on the positive-energy side (+E). The target side is not used to construct the ranking.

The operational question is therefore: can REAL identify the relevant target regions more efficiently and more accurately than a matched NULL control?

2. Frozen model and interpretation

Item	Frozen definition
Frozen Phase-3 kernel	v30.7 / namespace v30.2
System size	8 qubits
Hamiltonian	11-term random-Pauli model
Seed ensemble	25043000-25043011
Control	REAL versus matched NULL
Application endpoint	Cross-energy ranked search, -E -> +E
Primary target metric	Precision@8
Secondary metrics	Target evaluations and NDCG@8

Conceptual interpretation. The working application hypothesis is that Soft Spaces may act as a problem-adaptive prior over Hilbert space: rather than searching uniformly, the response structure identifies regions that should be inspected first.

3. v40.0 - Application screening

v40.0 tested whether the frozen Phase-3 structure can be converted into a measurable ranked-search advantage.

Metric	Result
Mean Precision@8 REAL	1.000000
Mean Precision@8 NULL	0.822917
Delta Precision@8	+0.177083
95% bootstrap CI	[+0.125000, +0.229167]
Sign test	p = 0.00048828

Wilcoxon	$p = 0.00097609$
Mean target evaluations REAL	8.000000
Mean target evaluations NULL	10.916667
Mean evaluations saved	2.916667
Mean NDCG@8 REAL	0.999940
Mean NDCG@8 NULL	0.897977
Decision	PASS

Interpretation: the REAL structure converts the Phase-3 correlation into an operational search advantage. REAL recovers the Top-8 set at essentially the theoretical minimum search effort, while NULL requires more target evaluations.

4. v40.1 - Controlled score-noise robustness

v40.1 kept the application endpoint fixed and corrupted only the source-side ranking information with controlled relative score noise.

Metric	Result
Primary checkpoint	$\eta = 1e-3$
Precision@8 REAL	0.998292
Precision@8 NULL	0.822917
Delta Precision@8	+0.175375
95% bootstrap CI	[+0.123292, +0.229167]
Sign test	$p = 0.00048828$
Wilcoxon	$p = 0.00097609$
Largest tested η passing all four gates	0.4
Decision	PASS

This result establishes robustness of the application signal against substantial corruption of the ranking information. It does not equate η with a physical gate-error probability.

5. v40.2 - Hardware-oriented proxy

v40.2 introduced finite shots, symmetric readout error and a depolarizing/circuit-depth proxy. Default parameters were 4096 shots, 1% readout error, $p_{2q} = 1.17e-3$, $p_{1q} = 1.17e-4$.

Metric	Result
Primary checkpoint	$N_{2q} = 20$
Precision@8 REAL	0.978021
Precision@8 NULL	0.825438
Delta Precision@8	+0.152583
95% bootstrap CI	[+0.102458, +0.207500]
Sign test	$p = 0.00048828$
Wilcoxon	$p = 0.00097609$
Largest tested depth passing all gates	$N_{2q} = 400$
Decision	PASS

v40.2 showed that the ranked-search advantage is not destroyed by a hardware-oriented proxy, but this remained a classical proxy rather than an explicit quantum-circuit simulation.

6. v40.3b - Explicit Qiskit/Aer circuit-level validation

v40.3b moved the source information through an explicit Qiskit/Aer measurement-channel circuit. The setup used Qiskit 2.5.2, Aer 0.17.2, Python 3.12.7, 4096 shots, $p_{1q} = 1.17e-4$, $p_{2q} = 1.17e-3$, 1% symmetric

readout error, and a deliberate CX identity-pair network to load two-qubit gate noise without changing the noiseless logical signal.

CX	REAL P@8	NULL P@8	Delta P	REAL eval	NULL eval	Decision
0	0.978021	0.825729	+0.152292	8.240	10.938	PASS
10	0.979427	0.825833	+0.153594	8.227	10.956	PASS
20	0.978333	0.825052	+0.153281	8.240	10.965	PASS
50	0.977031	0.825625	+0.151406	8.263	10.953	PASS
100	0.975990	0.826302	+0.149688	8.267	10.962	PASS
200	0.974844	0.825990	+0.148854	8.290	10.973	PASS
400	0.967292	0.825990	+0.141302	8.396	10.977	PASS

Primary checkpoint at CX=20. Delta Precision@8 = +0.153281, 95% bootstrap CI [+0.102448, +0.208281], sign-test p = 0.00048828 and Wilcoxon p = 0.00097609. Final decision: PASS.

The most important current result is that no breakdown was observed through CX=400. The REAL-specific advantage remained positive, statistically significant and operationally useful across the full tested circuit-noise range.

7. What has been established - and what has not

Status	Claim
Established	A reproducible REAL-specific ranked-search advantage exists in the frozen 8Q model.
Established	The advantage survives substantial score corruption.
Established	The advantage survives a finite-shot/readout/depolarizing hardware proxy.
Established	The advantage survives an explicit noisy Qiskit/Aer measurement-channel circuit through 400 CX gates.
Not yet established	The full Soft-Spaces Hamiltonian/projector calculation has not been implemented natively as a quantum algorithm.
Not yet established	The effect has not yet been demonstrated on IBM Heron or any other physical QPU.
Not yet established	No general claim has been established for arbitrary quantum programs or arbitrary Hamiltonian classes.

8. Current application interpretation

The emerging interpretation is a response-guided, problem-adaptive Hilbert-space architecture. Instead of selecting a predefined subspace first, the physical response structure is used to identify which regions should be prioritized.

In practical terms, the current evidence supports the statement that, for the tested model class, a Soft-Space-derived prior can reduce search effort and improve target ranking relative to a matched NULL reference.

9. Immediate next experiment: v40.3c breakpoint scan

Because the full v40.3b run is computationally expensive and no breakdown was found through 400 CX gates, the next step is a deliberately reduced exploratory scan designed only to locate the approximate breakdown region.

Parameter	v40.3c
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Seeds	First 4 frozen seeds
Repetitions	50 per seed/model/CX
Shots	1024
CX values	800, 1200, 2000, 4000
Purpose	Locate the first weak or near-breakdown region
Scientific status	Exploratory screening only; not confirmatory

10. Route toward IBM Heron

If the breakpoint scan shows that the REAL-specific advantage persists well beyond the operating regime relevant to the planned circuit, the next meaningful validation is execution on a physical IBM QPU. A conservative first hardware experiment would use an 8-qubit subgraph and compare the same REAL versus NULL ranked-search endpoint under real device calibration, gate errors, decoherence and readout errors.

A stronger replication would repeat the experiment on more than one Heron processor to test whether the advantage is reproducible across devices rather than being specific to one chip or calibration.

11. Overall Phase 4 status

v40.0 PASS -> v40.1 PASS -> v40.2 PASS -> v40.3b PASS

Phase 4 has therefore crossed an important threshold: the Soft-Spaces result is no longer only a structural observation. Within the frozen 8Q model, it now has a measurable application-level consequence that remains robust through an explicit noisy circuit simulation. The next scientific tasks are to locate the circuit-noise breakdown region and then test the same operational claim on physical quantum hardware.